

Кладковой Максим 5130904/30002 Вар 47

Ортогонализация гауссовых методом Грама-Шмидта

$$f(x) = \sin(\pi x); x \in [-\frac{1}{4}, \frac{1}{4}]; n=2 \quad Q_2(x) = \beta_0 \psi_0 + \beta_1 \psi_1 + \beta_2 \psi_2$$

$$\psi_0(x) = 1, \quad \psi_1(x) = x, \quad \psi_2(x) = x^2$$

• Вычисление ψ_0 :

$$\tilde{\psi}_0 = 1; \quad \|\psi_0\|_2 = \sqrt{\int_{-\frac{1}{4}}^{\frac{1}{4}} dx} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{1}{2}}; \quad \psi_0 = \frac{1}{\sqrt{\frac{1}{2}}} = \sqrt{2}$$

$$\tilde{\psi}_1 = \psi_1 - r_{01} \psi_0; \quad \tilde{\psi}_1 = x - r_{01} \sqrt{2}$$

$$r_{01} = (\psi_1, \psi_0) = \int_{-\frac{1}{4}}^{\frac{1}{4}} x \sqrt{2} dx = 0; \quad \tilde{\psi}_1 = x$$

$$\|\psi_1\|_2 = \sqrt{\int_{-\frac{1}{4}}^{\frac{1}{4}} x^2 dx} = \sqrt{\frac{x^3}{3} \Big|_{-\frac{1}{4}}^{\frac{1}{4}}} = \sqrt{\frac{1}{3} \left(\frac{1}{64} + \frac{1}{64} \right)} = \sqrt{\frac{1}{96}}; \quad \psi_1 = \frac{\tilde{\psi}_1}{\|\psi_1\|_2} = \frac{x}{\sqrt{\frac{1}{96}}} = x \sqrt{96}$$

$$\tilde{\psi}_2 = \psi_2 - r_{02} \psi_0 - r_{12} \psi_1; \quad r_{02} = (\psi_2, \psi_0) = \int_{-\frac{1}{4}}^{\frac{1}{4}} x^2 \sqrt{2} dx = \frac{x^3 \sqrt{2}}{3} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = \frac{\sqrt{2}}{96}$$

$$r_{12} = (\psi_2, \psi_1) = \int_{-\frac{1}{4}}^{\frac{1}{4}} x^2 \cdot x \sqrt{96} dx = \frac{x^4 \sqrt{96}}{4} \Big|_{-\frac{1}{4}}^{\frac{1}{4}} = 0; \quad \tilde{\psi}_2 = x^2 - \frac{1}{48}$$

$$\|\psi_2\|_2 = \sqrt{\int_{-\frac{1}{4}}^{\frac{1}{4}} (x^2 - \frac{1}{48})^2 dx} = \sqrt{\left(\frac{x^5}{5} - \frac{x^3}{72} + \frac{1}{2304} \right) \Big|_{-\frac{1}{4}}^{\frac{1}{4}}} = \sqrt{\frac{1}{5760}}; \quad \psi_2 = \frac{x^2 - \frac{1}{48}}{\sqrt{\frac{1}{5760}}} = 24\sqrt{10}x^2 - \frac{\sqrt{10}}{2}$$

• Вычисление β :

$$\beta_0 = (f, \psi_0) = \int_{-\frac{1}{4}}^{\frac{1}{4}} \sin(\pi x) \sqrt{2} dx = 0$$

$$\beta_1 = (f, \psi_1) = \int_{-\frac{1}{4}}^{\frac{1}{4}} \sin(\pi x) \cdot x \sqrt{96} dx = -\frac{2\sqrt{3}}{\pi} + \frac{8\sqrt{3}}{\pi^2} \approx 0,301$$

$$\beta_2 = (f, \psi_2) = \int_{-\frac{1}{4}}^{\frac{1}{4}} (24\sqrt{10}x^2 - \frac{\sqrt{10}}{2}) \sin(\pi x) dx = 0$$

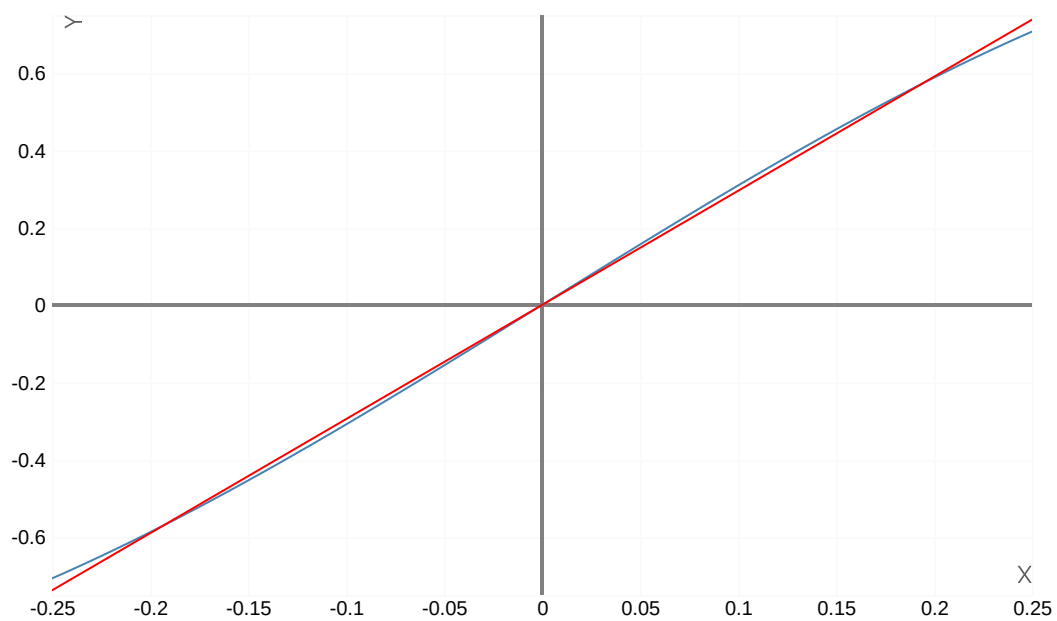
$$Q_2(x) = \beta_1 \psi_1 = 0,301 \cdot x \sqrt{96} \approx 2,952x$$

$$\varepsilon(x) = \sin(\pi x) - 2,952x$$

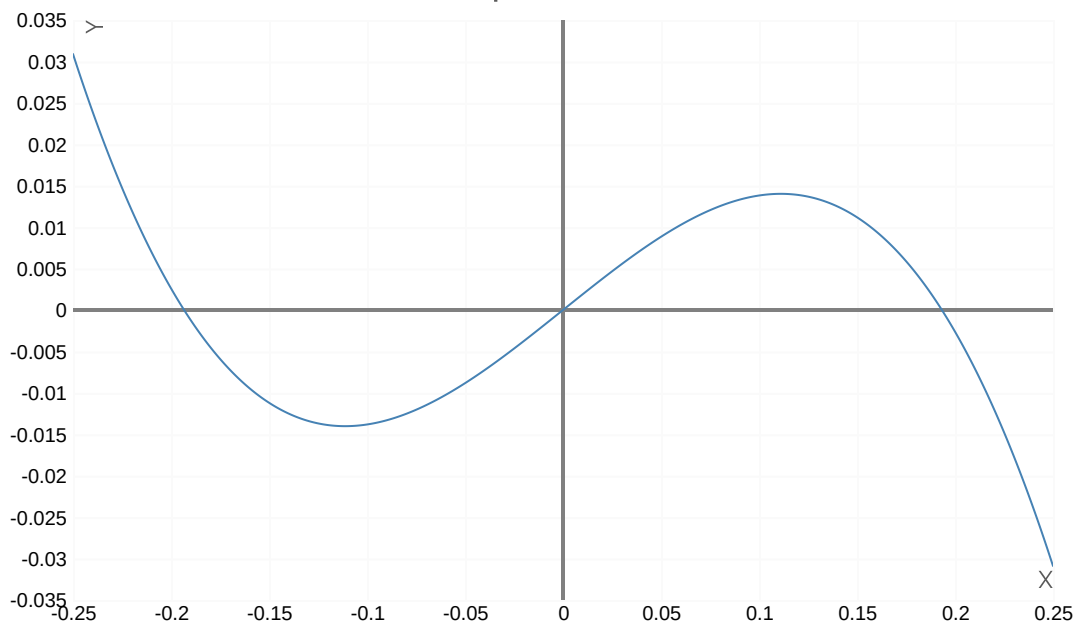
$$\varepsilon^2(x) \approx \sin^2(\pi x) - \frac{738x \sin \pi x}{125} + 8,714x^2$$

$$\rho = 2 \int_{-\frac{1}{4}}^{\frac{1}{4}} \sin^2(\pi x) - \frac{738x \sin(\pi x)}{125} + 8,714x^2 \approx 1,3 \cdot 10^{-4}$$

$\sin(\pi \cdot x); \mathbb{Q}[2]x$



$\epsilon(x)$



$$\epsilon(x)^2$$

